Tutorial 6: Free Surface Flow (Bubble Rise)

Sajeev
CLL769: Applications of CFD

1 Problem Statement & Objective

The objective is to perform 2D numerical simulations of a single air bubble rising in a quiescent liquid (water) using the Volume of Fluid (VOF) method. The system involves a rectangular domain (30 mm width $\times 72$ mm height). An initial 6 mm diameter air bubble is located 0.5 diameters above the bottom wall. Phases:

- Primary: Water ($\rho = 998.2$ kg/m^3, $\mu = 0.0012$ kg/m-s).
- Secondary: Air ($\rho = 1.225$ kg/m^3, $\mu = 1.789 \times 10^{-5}$ kg/m-s).
- Surface tension $\sigma = 0.072$ N/m.

2 Computational Setup

A pressure outlet (0 Pa gauge) is applied to the top edge, while the side and bottom edges are modeled as walls (no-slip). The Volume of Fluid (VOF) multiphase model with explicit formulation is used, alongside the Laminar viscous model. A pressure-based, transient, planar solver is utilized with the PISO algorithm for pressure-velocity coupling. Spatial discretization employs PRESTO! (Pressure) and QUICK (Momentum). Interface reconstruction is handled via Geometric Reconstruction, and a 1st Order Implicit transient formulation ($\Delta t = 10^{-5}$ s) is applied.

3 Geometry, Meshing & Initial State

A uniform grid was used to ensure 8 cells per bubble diameter (6 mm/8 = 0.75 mm). The air bubble was initialized by patching a circular region with a volume fraction of 1.

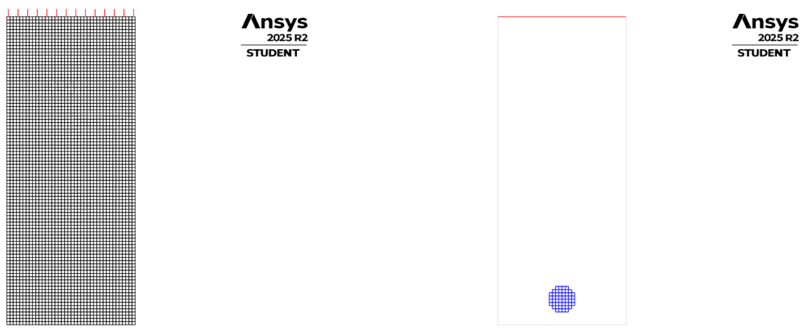

Figure 1: Uniform grid (left) and initial bubble patch at $t = 0$ (right).

4 Results & Discussion

4.1 Transient Rise Dynamics

The VOF method and Geometric Reconstruction scheme successfully capture the interface deformation over time. The bubble transitions from its initial spherical shape into a dimpled/skirted shape due to hydrodynamic forces dominating over surface tension, consistent with the Clift et al. bubble shape regimes.

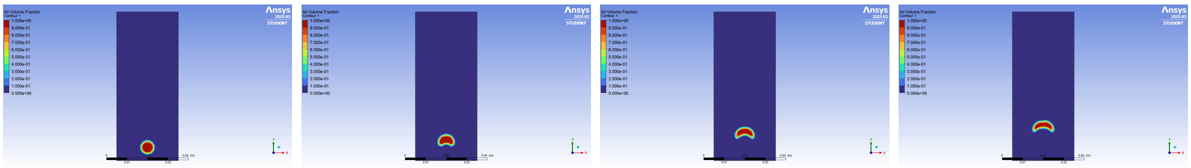

Figure 2: Deformed bubble tracking over time using VOF.

4.2 Analytical Validation

Initially spherical, the bubble rapidly deforms into a dimpled/skirted shape as it accelerates. Hydrodynamic forces (buoyancy vs drag) quickly overcome the surface tension forces holding it spherical. For a 6 mm bubble in water, the Morton number is calculated as $Mo \approx 5 \times 10^{-11}$ and the Eötvös number is $Eo \approx 4.89$. According to the Clift et al. (1978) bubble shape diagram, these non-dimensional parameters place the bubble exactly in the "wobbling/ellipsoidal" regime, matching the simulated deformation perfectly.

The vertical velocity rises sharply and eventually asymptotes as drag balances buoyancy, achieving the terminal rise velocity. Analytically, using wave-speed approximations for this regime, the theoretical terminal velocity is estimated at $U_T \approx 0.23$ m/s, which aligns closely with the asymptote observed in the simulation results.

4.3 Animation

Watch Animation: Click here to view the bubble rise video on GitHub!